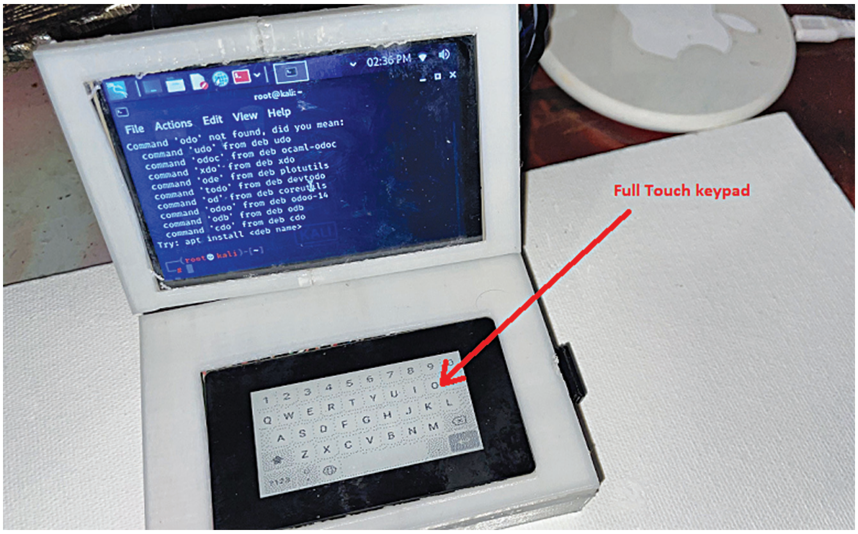

Fig. 5: The laptop's touch keyboard

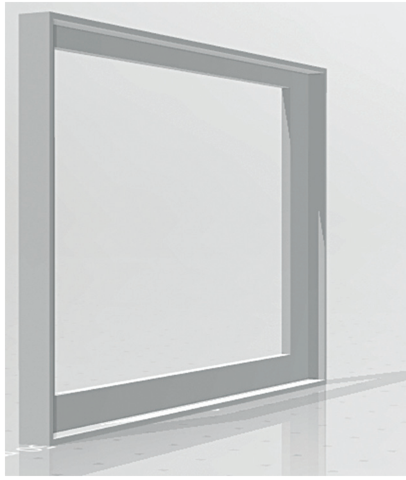

Fig. 8: LCD panel

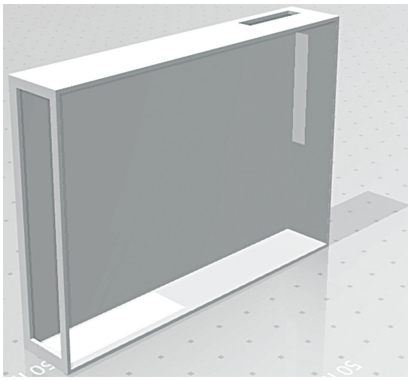

Fig. 6: The laptop's base

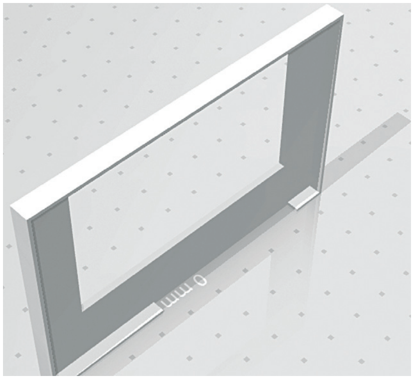

Fig. 7: Base cover to hold the keyboard

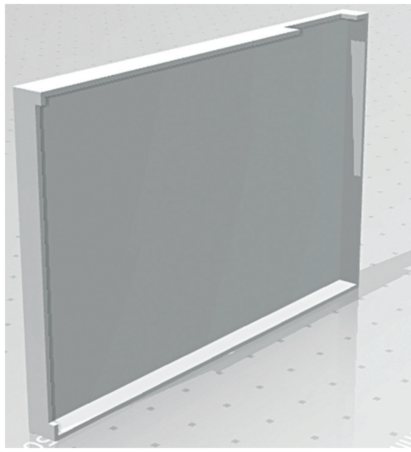

Fig. 9: LCD panel's back cover

ploitation tools for penetration hacking and testing, such as Metasploit, Aircrack-ng, and Wireshark.

Designing the laptop

After collecting the components listed in the Bill of Materials, we need to design the case or body of the laptop that is to hold the hardware inside it safely. We may design the laptop body in three parts. First, the base part where we keep the main hardware, viz, Raspberry Pi, battery, and charging system. In the base we make two cuts: one in front for the Ethernet and USB and the other for power and charging of the battery.

Next, we design the top cover of the base to hold a touch e-paper based keyboard. We make a cut in the top cover to fix the touch panel keyboard inside it, as shown in Fig. 7.

Next, we design the LCD panel's support for the laptop. For that we make the enclosure in the same size as the display and keep the thickness same as well.

After designing the laptop body, we can 3D print its parts.

Prerequisite

Before proceeding further, we need to prepare the Raspberry Pi board with Kali Linux. For that, go to Kali Linux official website and download its latest version for an ARM based device like Raspberry Pi. Next, download the Raspberry Pi disk imager and install it. Insert the SD card in USB drive and prepare it with Kali Linux OS using Raspberry Pi disk imager by choosing Kali Linux OS image file in Raspberry Pi disk imager.

Preparing OS for SPI display

For our laptop we can choose any display for Raspberry Pi that is up to 8.9cm (3.5-inch) in size. An HDMI based touch display is recommended as we are using a touch e-ink display for the keyboard panel, which is an SPI, and two SPI displays can cause conflict in communication. However, the SPI touch e-ink display used for keyboard uses only its touch function that is I2C based, so the SPI of other displays does not cause that much of a problem.

An SPI based display was used in the prototype because it had already been bought, but use of 8.9cm HDMI touch display is recommended in-